

CONVOLUTIONAL NEURAL NETWORK

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Great visualization tool:
<https://poloclub.github.io/cnn-explainer/>

Facial Recognition

Deep Learning Neural Networks use layers of increasingly complex rules to categorize Complicated shapes such as faces

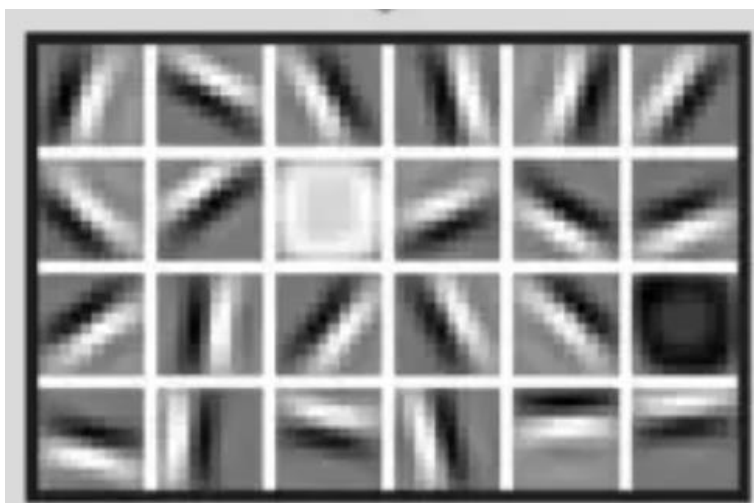
Layer 1: The model identifies pixels of light and dark



Layer 3: The model learns to identify more complex shapes and objects



Layer 2: The model learns to identify edges and simple shapes

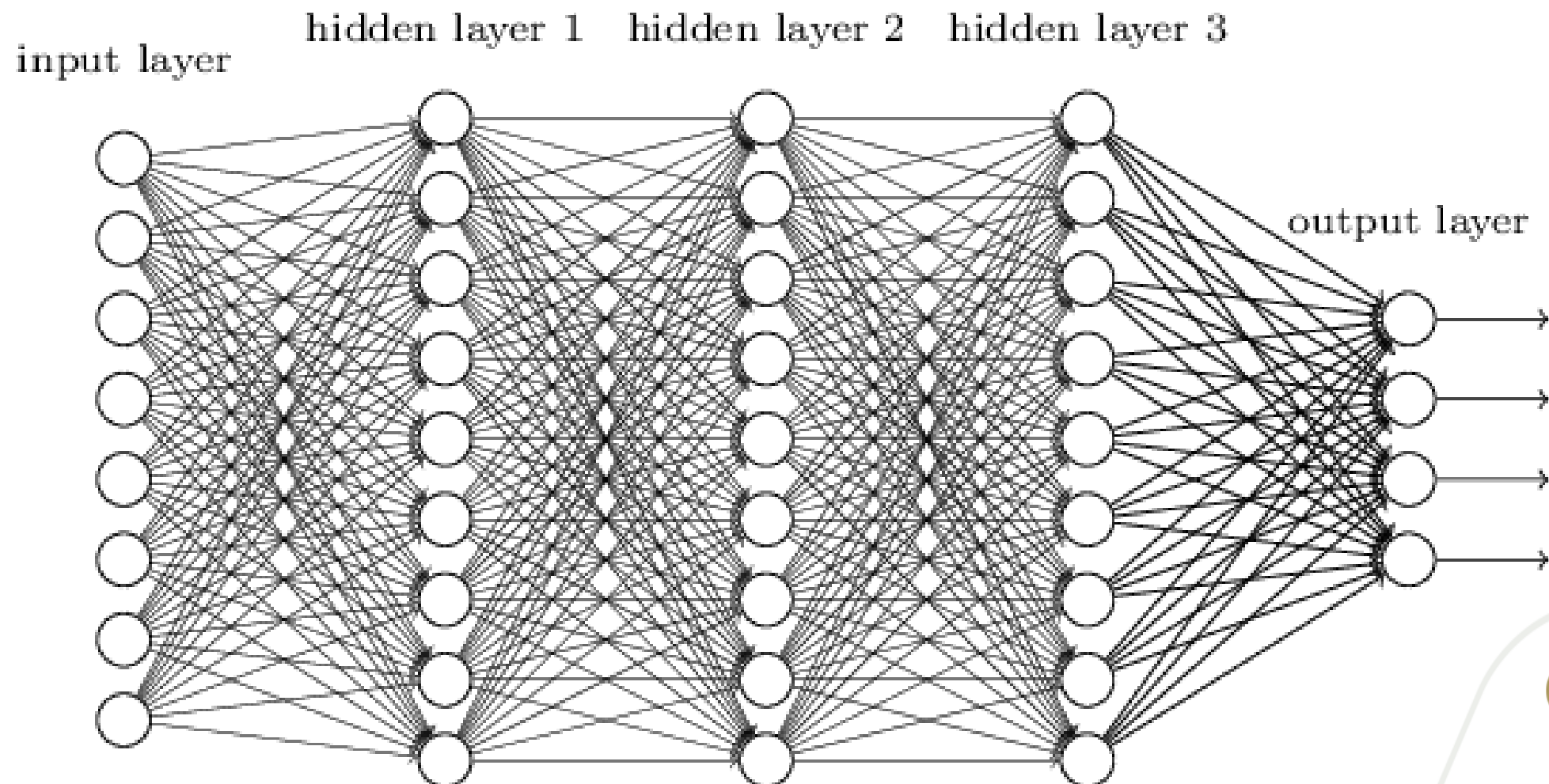


Layer 4: The model learns which shapes and objects can be used to define a human face



Smaller Network: CNN

- We know it is good to learn a small model.
- From this fully connected model, do we really need all the edges?
- Can some of these be shared?



1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

1	3	2	6	0	1
0	1	0	0	1	0
0	7	1	1	0	0
1	0	4	5	1	8
4	1	0	7	2	0
3	0	1	0	1	0



0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0



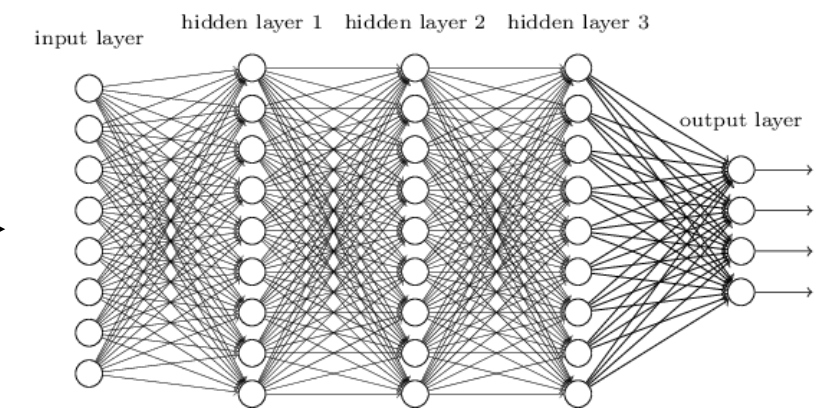
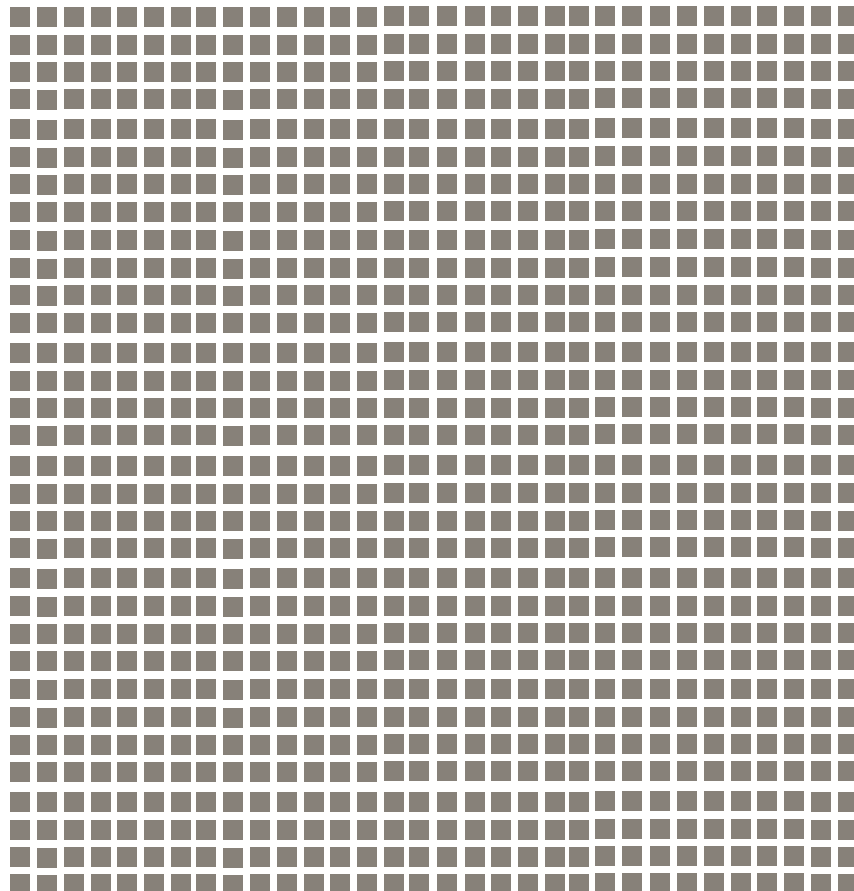
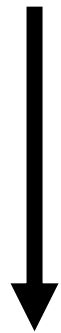
Consider learning an image:

- Some patterns are much smaller than the whole image

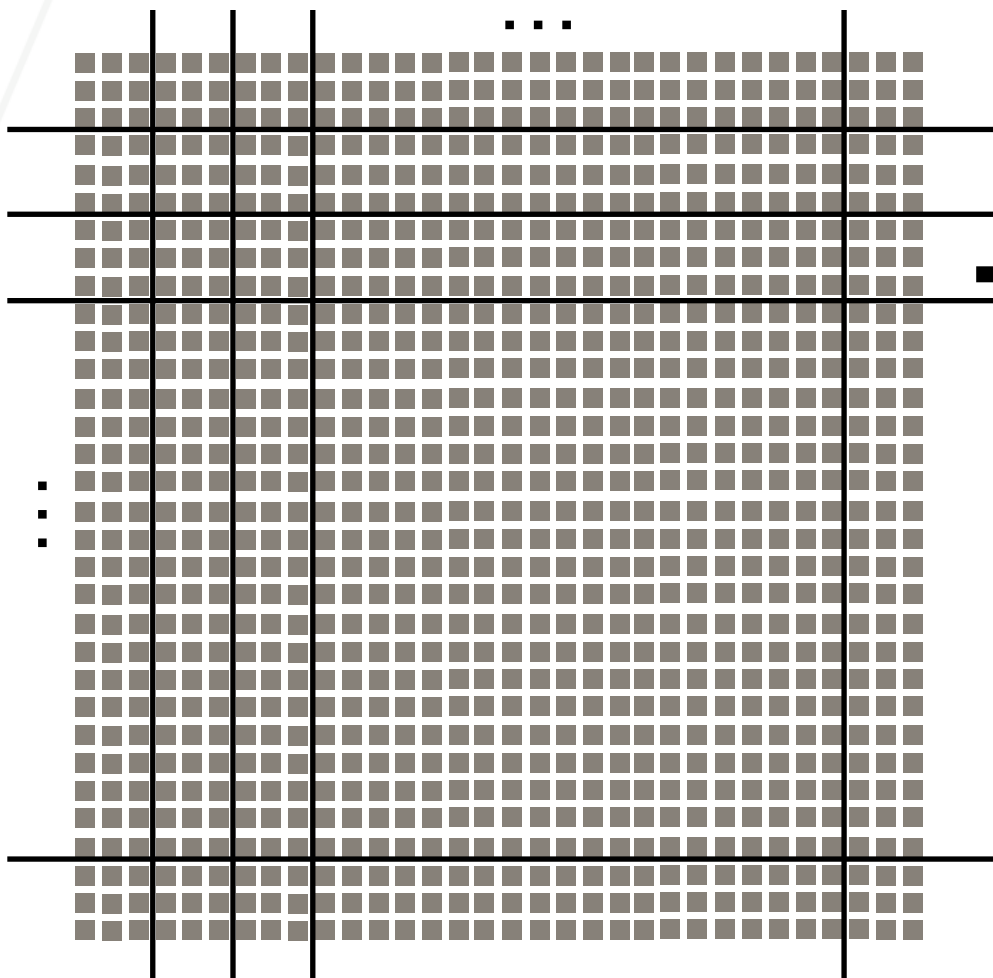
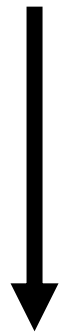
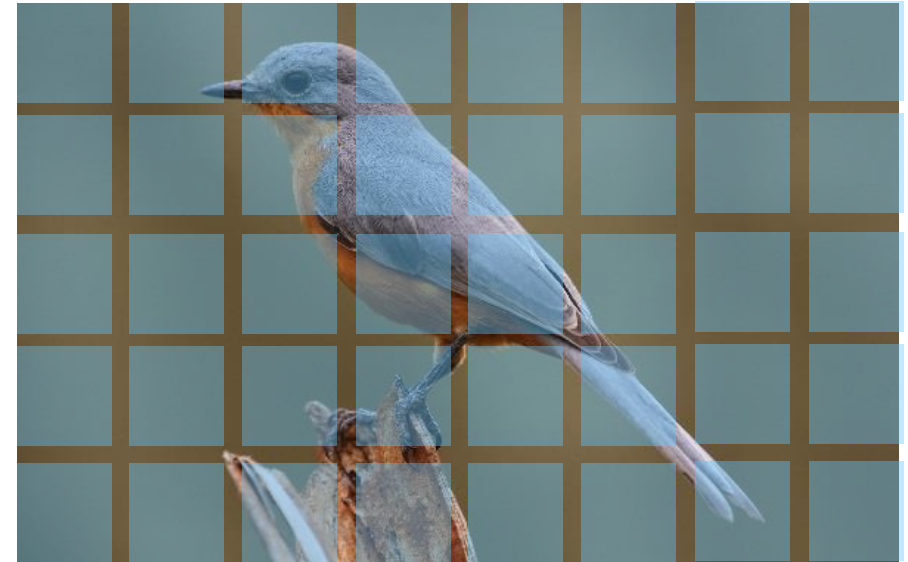
Can represent a small region with fewer parameters



600x600



600x600



3x3



3x3

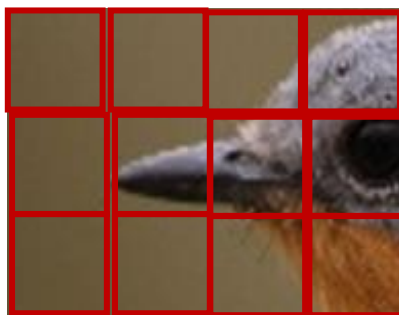


3x3

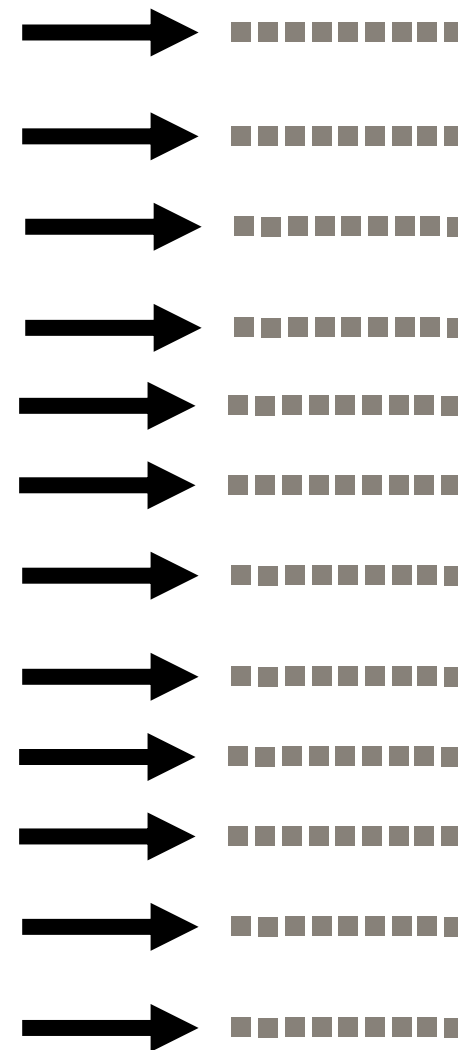
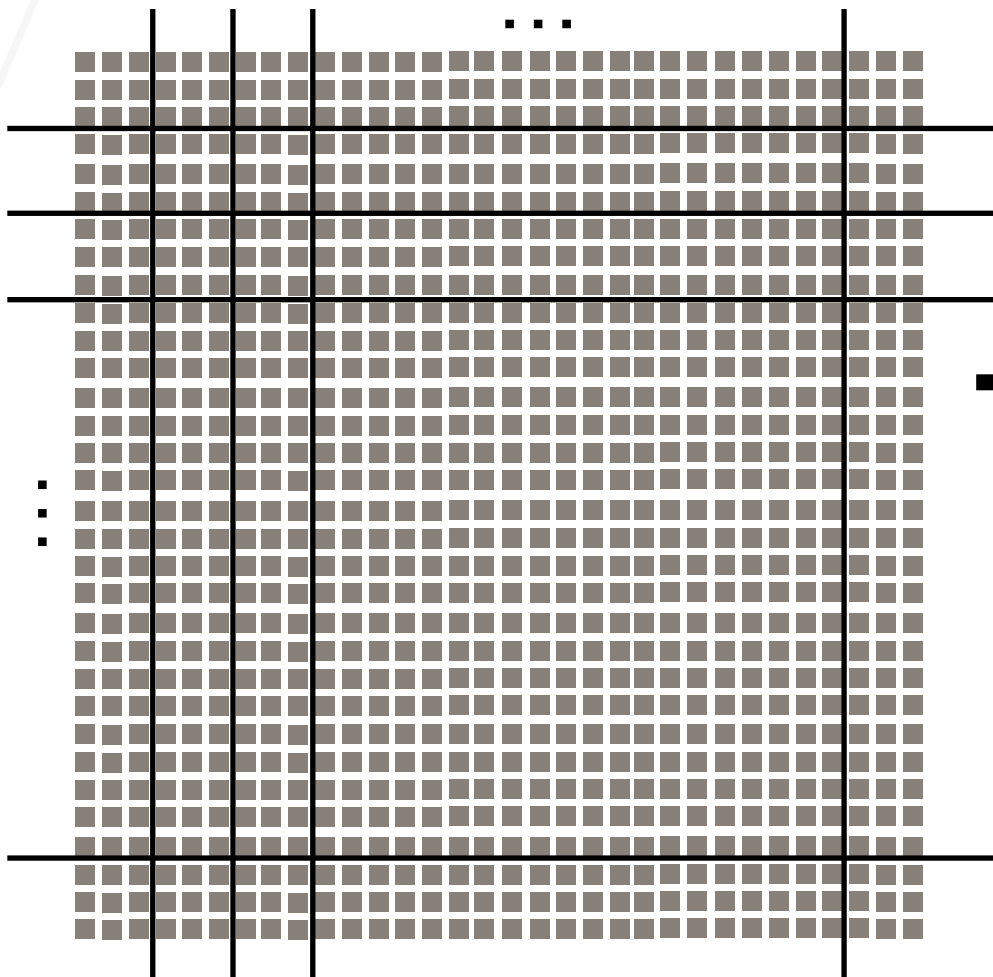


⋮

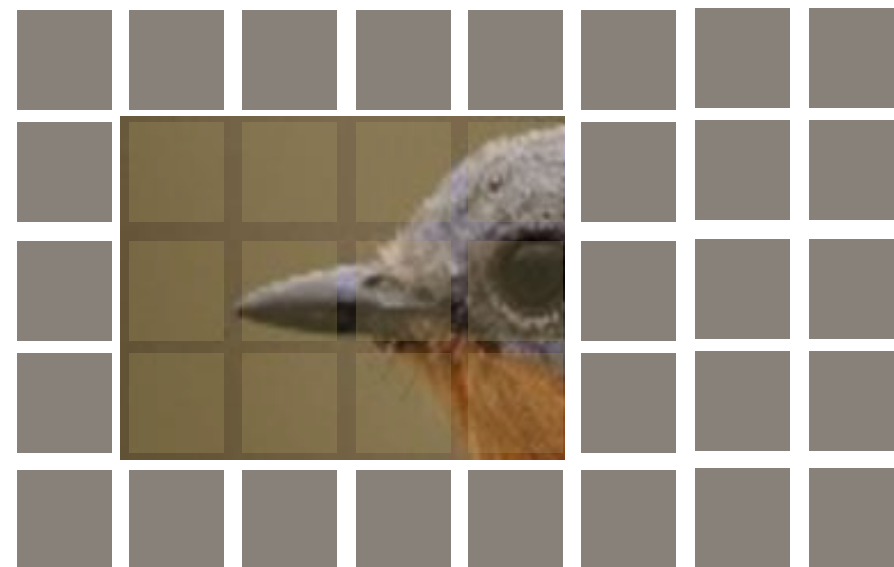




9x12



...



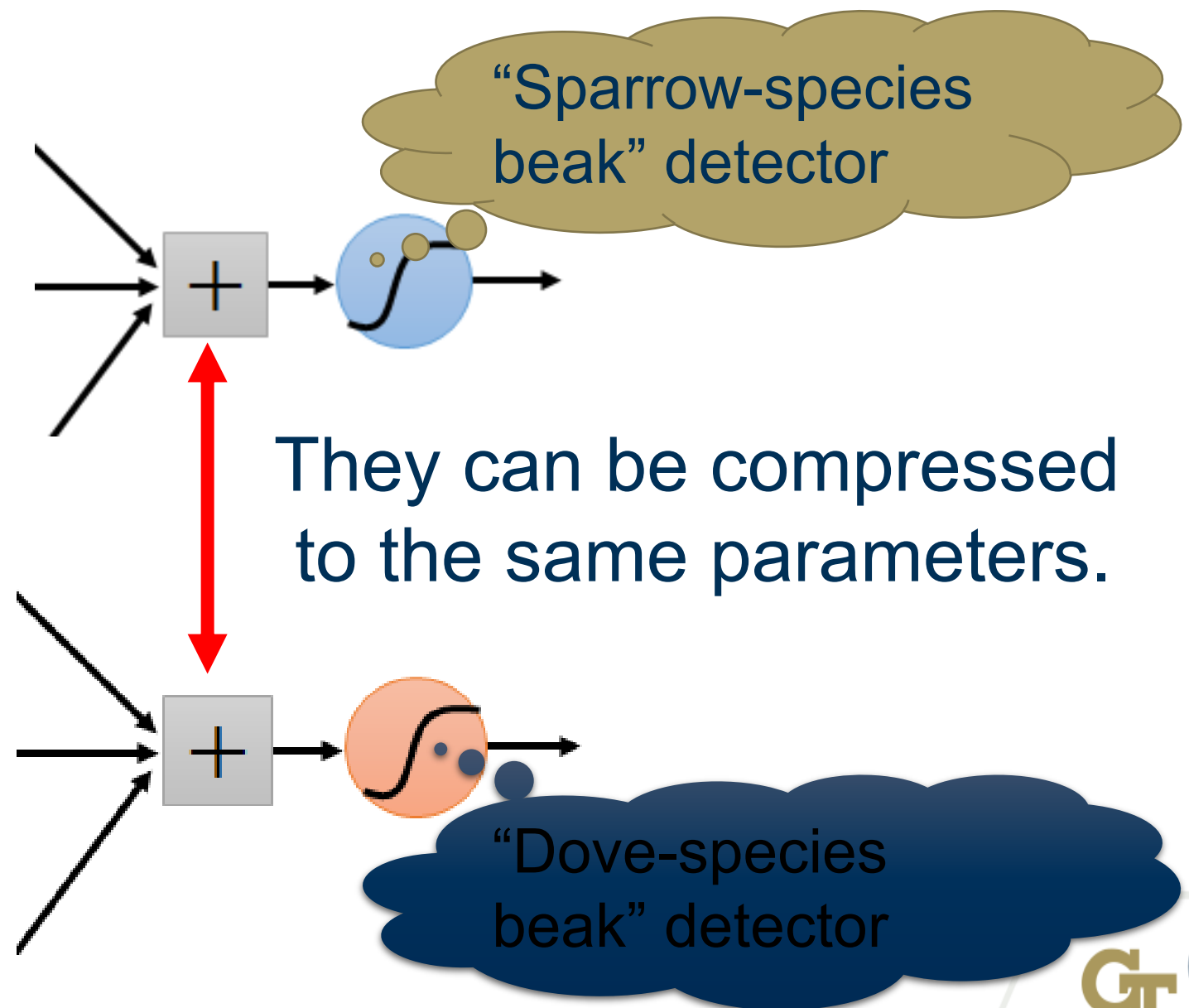
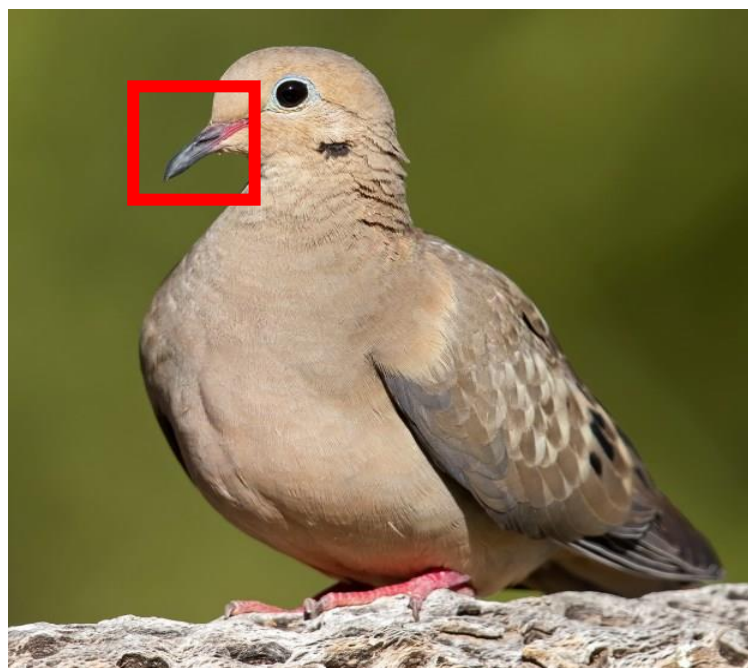
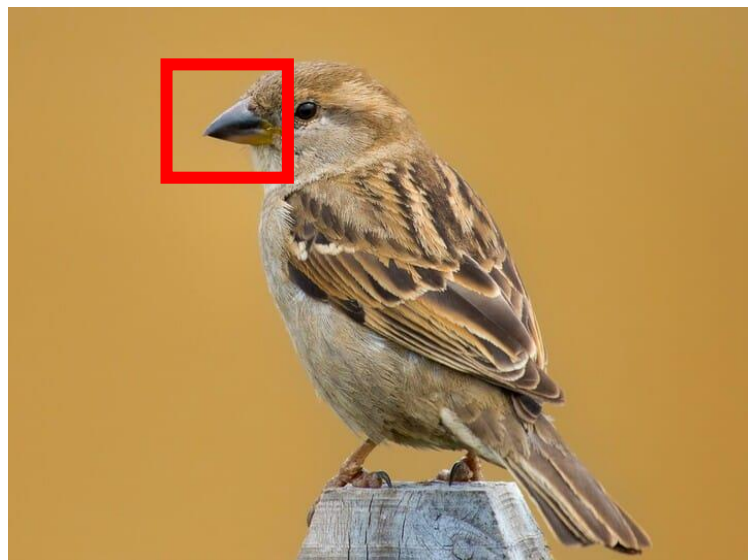
...

Why is NN wasteful?

- Each neuron in a fully connected network “sees” *every pixel*, no matter where it is in the image.
But that’s unnecessary for images — because important features like **beaks**, **wings**, **eyes** tend to be **local** and **repeat** in different places.
- So, in a standard NN, you’d need to *independently learn a beak detector* for:
 - every position (top-left, center, bottom-right),
 - every bird,
 - every image.
- Each location would require **its own set of weights**, hence *millions* of parameters.

Same pattern appears in different places:
They can be compressed!

**What about training a lot of such “small” detectors
and each detector must “move around”.**



How CNNs fix this with parameter sharing

CNNs say:

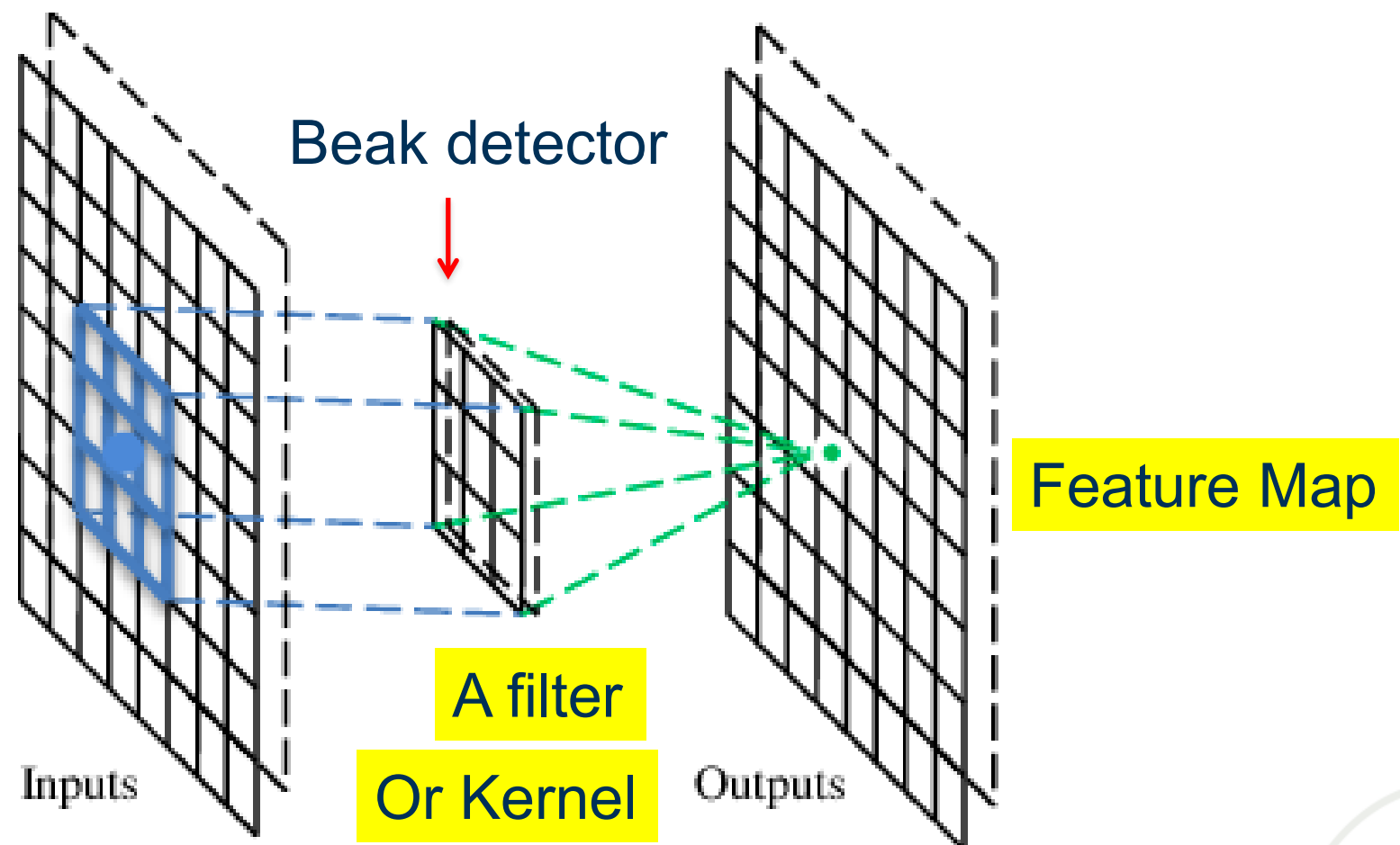
- “Instead of learning a separate detector for every position, let’s **use one small filter** (like a 3×3 beak pattern) and **slide it** over the whole image.”

Now:

- You only need **9 weights** (for a 3×3 filter),
- And you reuse them across *all locations*.
- That reuse — the **same weights everywhere** — is **parameter sharing**.

A convolutional layer

A CNN is a neural network with some convolutional layers (and some other layers). A convolutional layer has a number of filters that does convolutional operation.



Convolution

We want to detect vertical lines and diagonal lines in the binary image

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

These are the network parameters to be learned.

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

⋮ ⋮

Each filter detects a small pattern (3 x 3).

Convolution

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

Dot
product



3

-1

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

Convolution

If stride=2

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

3

-3

Convolution

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

Convolution

stride=1

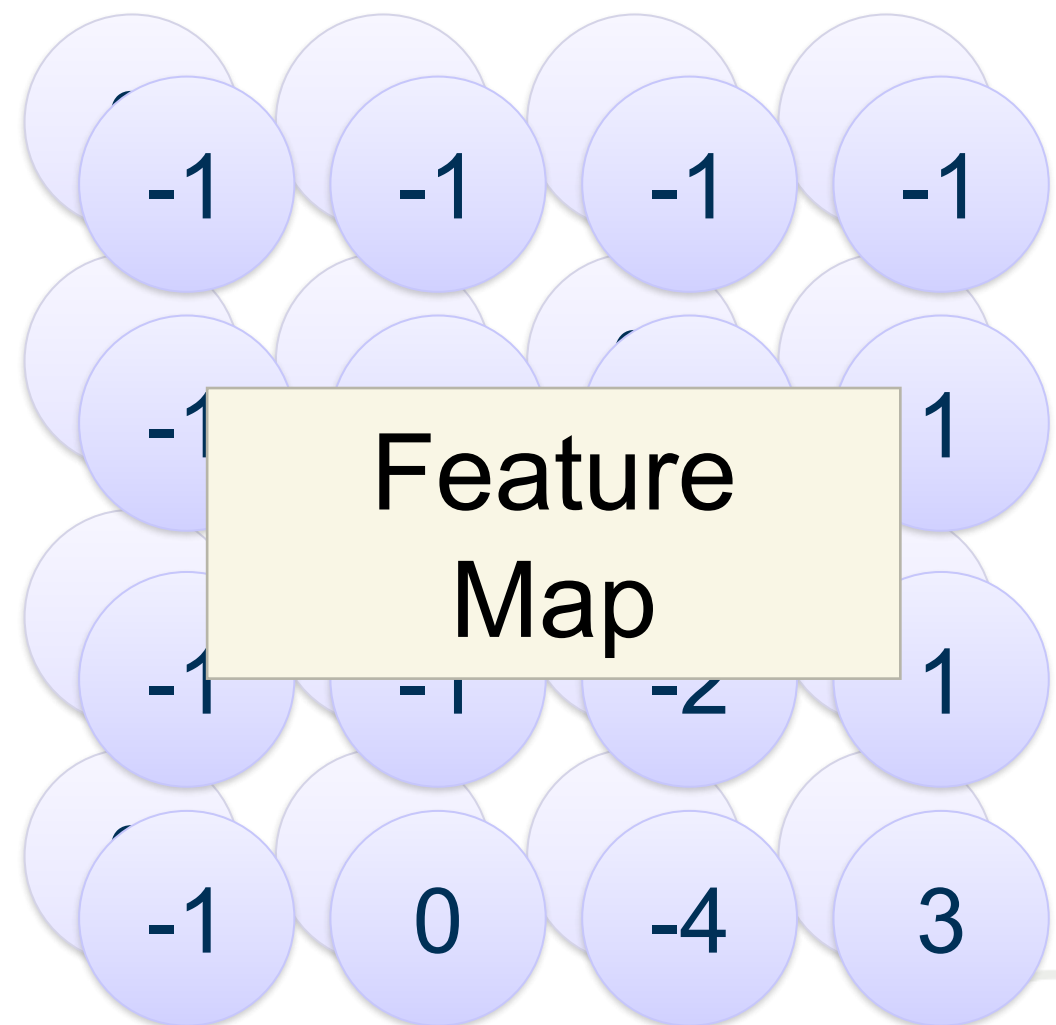
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

Repeat this for each filter



Two 4 x 4 images
Forming 2 x 4 x 4 matrix

How Convolution Learns to Detect Features

① Local Feature Detection

- Each **filter** in a CNN acts as a *small pattern detector*.
- A filter that “looks like” a **beak shape** will produce **high activation** wherever a similar pattern appears in the image.
- In the feature map, these high activations mark **where the beak is located**.

② Parameter Sharing Across Locations

- The same learned filter slides across the image — this means the **same weights** detect the same pattern *anywhere*.
- A filter trained to recognize a **sparrow’s beak** will also respond to a **dove’s beak** if the shape is similar.
- This ability to reuse the same detector across different positions is what makes CNNs **efficient and translation-invariant**.
-

• ③ Multiple Filters, Multiple Features

- CNNs learn **many filters**, each detecting a different feature (edges, textures, beaks, wings, eyes, etc.).
- These feature maps stack together, allowing the network to recognize **multiple parts of an object** simultaneously.
- Higher layers then combine these local detections to recognize **entire objects or categories** (e.g., “bird”).

Color image: RGB 3 channels

Color image

Filter 1

Filter 2

The diagram illustrates the process of converting a color image to grayscale using two filters. The color image is represented as three stacked planes (Red, Green, Blue). These are processed by Filter 1 and Filter 2, which are 3x3 matrices. The results are then combined into a single grayscale output.

Filter 1 matrix:

1	-1	-1
-1	1	-1
-1	-1	1

Filter 2 matrix:

-1	1	-1
-1	1	-1
-1	1	-1

The resulting grayscale image is shown as a single plane, which is the combination of the filtered color channels.

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Color image: RGB 3 channels

Color image

Filter 1

Filter 2

1 0 0 0 0 1

0 1 0 0 1 0

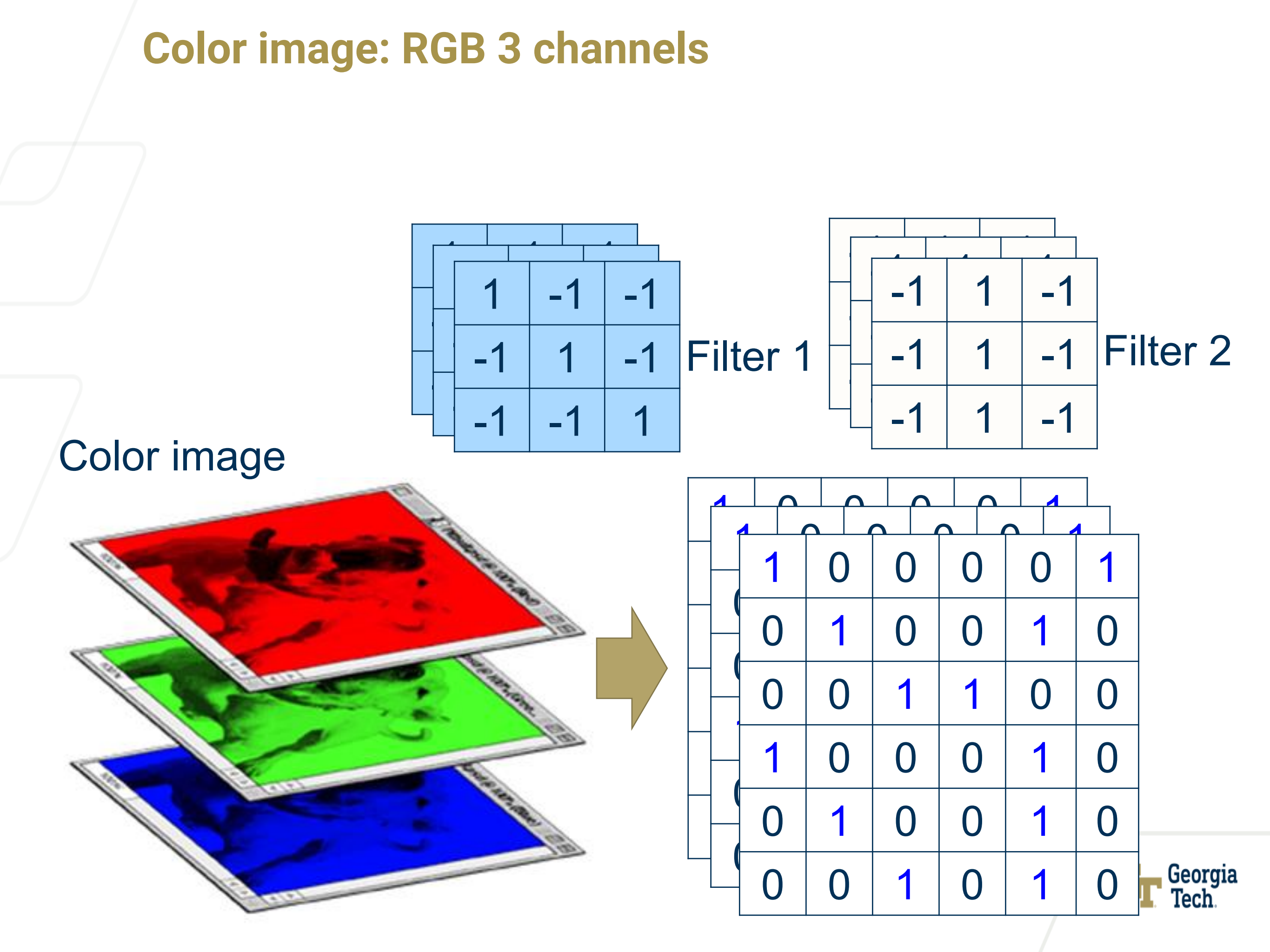
0 0 1 1 0 0

1 0 0 0 1 0

0 1 0 0 1 0

0 0 1 0 1 0

Georgia Tech.



Color image: RGB 3 channels

Color image

Filter 1

Filter 2

1 0 0 0 0 1

0 1 0 0 1 0

0 0 1 1 0 0

1 0 0 0 1 0

0 1 0 0 1 0

0 0 1 0 1 0

Georgia Tech.

Color image: RGB 3 channels

Color image

Filter 1

Filter 2

1 0 0 0 0 1

0 1 0 0 1 0

0 0 1 1 0 0

1 0 0 0 1 0

0 1 0 0 1 0

0 0 1 0 1 0

Georgia Tech.

Color image: RGB 3 channels

Color image

Filter 1

Filter 2

1 0 0 0 0 1

0 1 0 0 1 0

0 0 1 1 0 0

1 0 0 0 1 0

0 1 0 0 1 0

0 0 1 0 1 0

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Quick Knowledge Check

- During backpropagation, why do we apply the chain rule? A. To move from input to output B. To compute derivatives layer by layer C. To skip intermediate layers D. To reset gradients
- If the gradient of a weight is positive, how should the weight change in gradient descent? A. Decrease the weight B. Increase the weight C. Keep it constant D. Randomly perturb it.
- What is one practical way to mitigate exploding gradients? A. Using smaller datasets B. Gradient clipping C. Removing activation functions D. Increasing batch size
- What does parameter sharing mean in CNNs? A. Each neuron has unique weights B. Different filters are random C. The same filter (set of weights) slides across all image locations D. Parameters are averaged
- What is the effect of using multiple filters in a convolutional layer? A. Network becomes shallower B. Filters average pixels C. It limits generalization. D. Each filter learns to detect a different visual pattern (edges, textures, shapes)
- Compared to a fully connected layer, a convolutional layer: A. Requires more parameters B. Uses local connections and shared weights C. Connects each neuron to every pixel D. Ignores spatial structure

Convolution v.s. Fully Connected

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

image

1	-1	-1
-1	1	-1
-1	-1	1

-1	1	-1
-1	1	-1
-1	1	-1

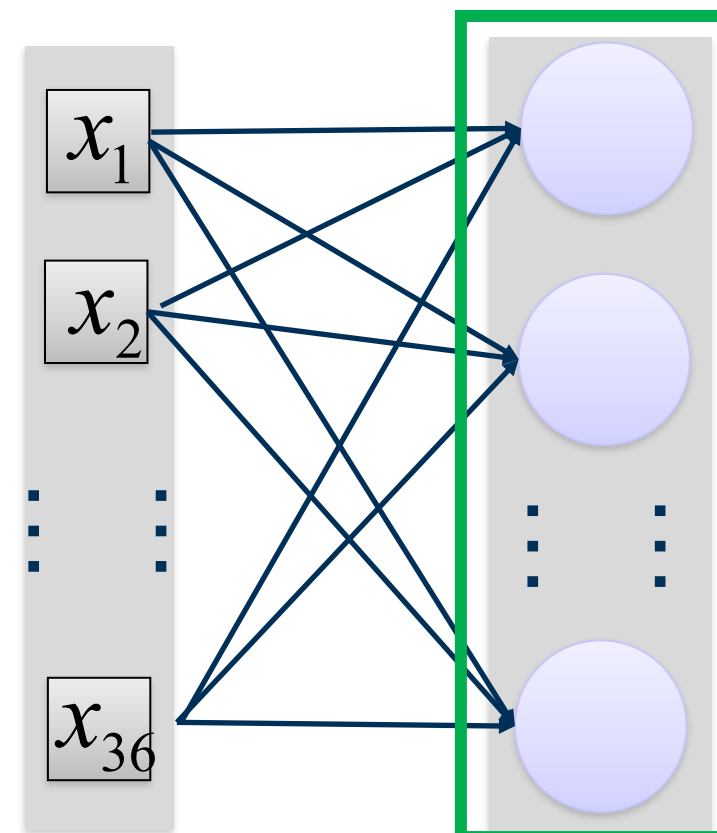


convolution

-1	-1	-1	-1
-1	-1	-2	1
-1	-1	-2	1
-1	0	-4	3

Fully-
connected

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0



Conventional Fully Connected layers (FC layers)

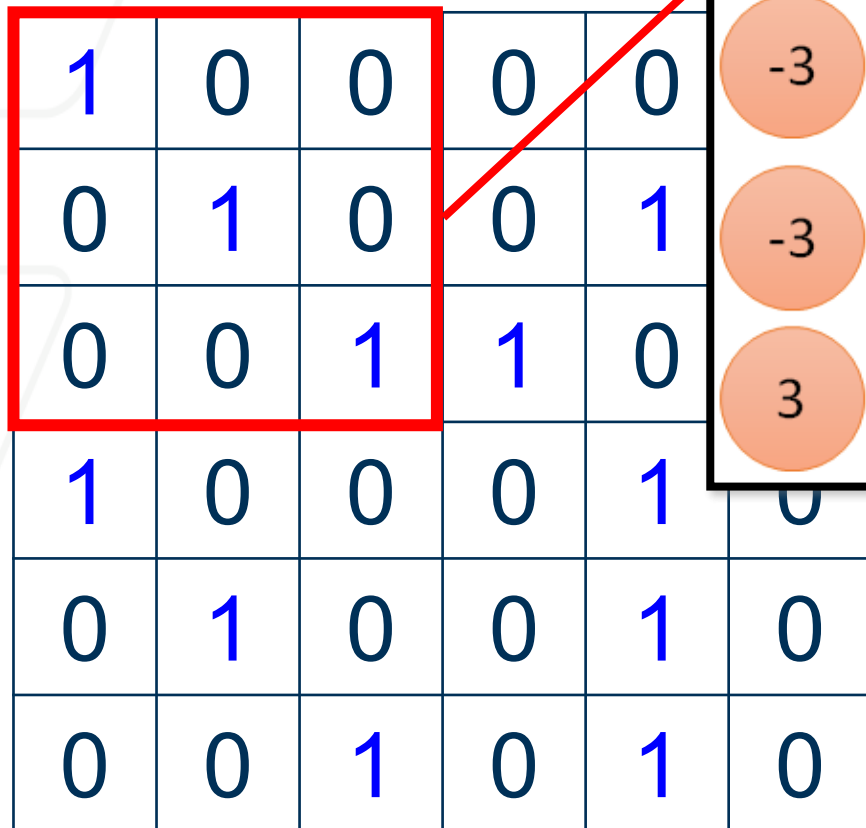
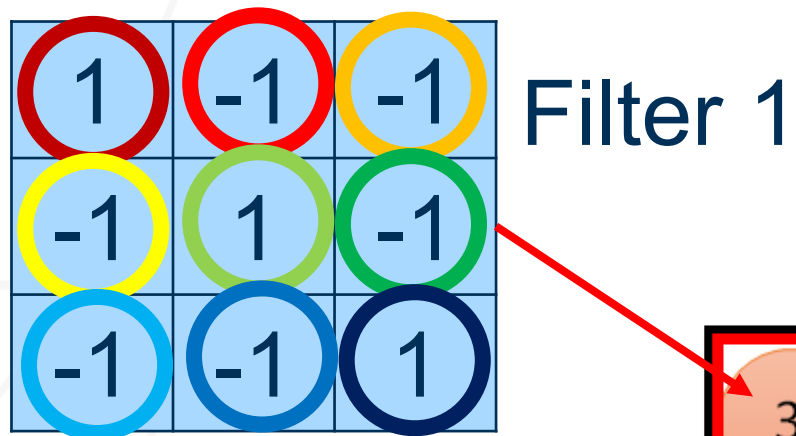
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	1	
2	0	
3	0	
4	0	
5	0	
6	1	
7	0	
8	1	
⋮		⋮
31	0	
32	0	
33	1	
34	0	
35	1	
36	0	

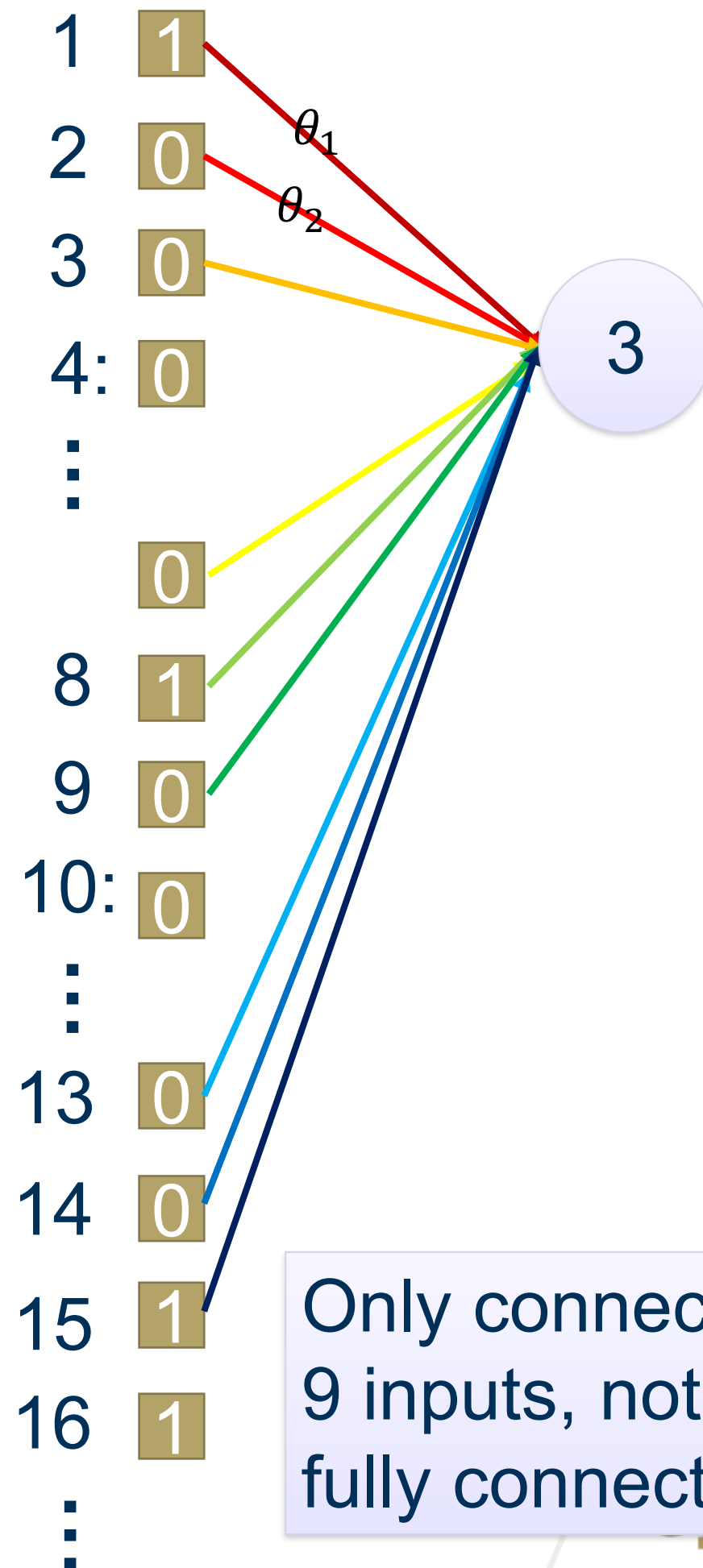
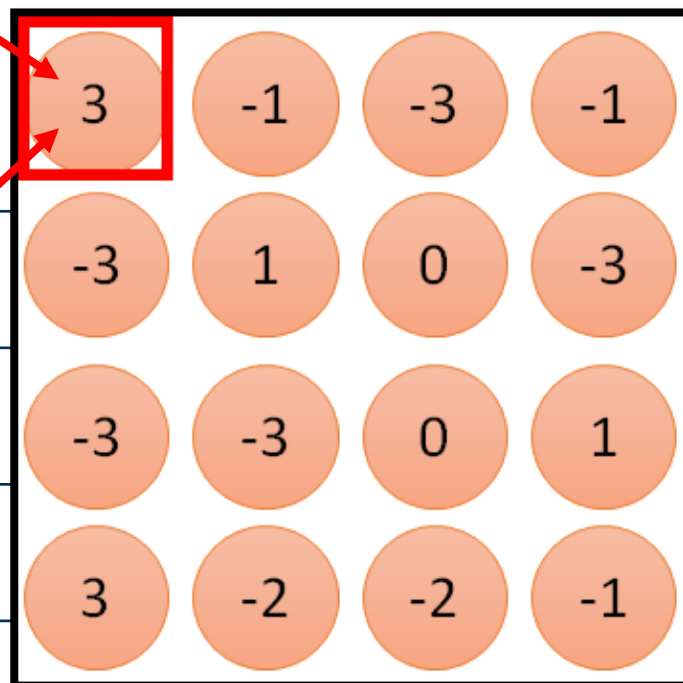
...

features 1st hidden layer

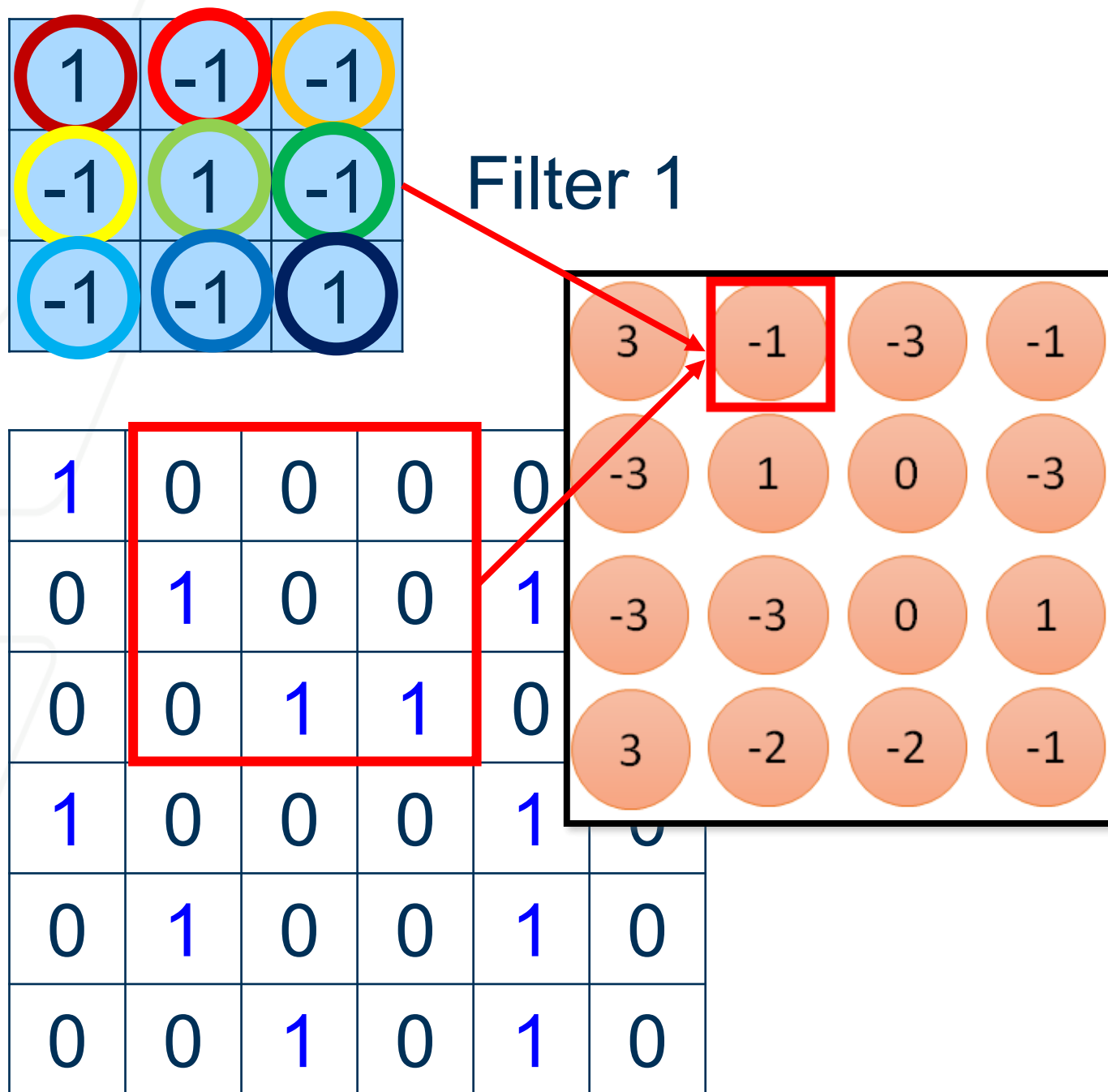


6 x 6 image

fewer parameters!



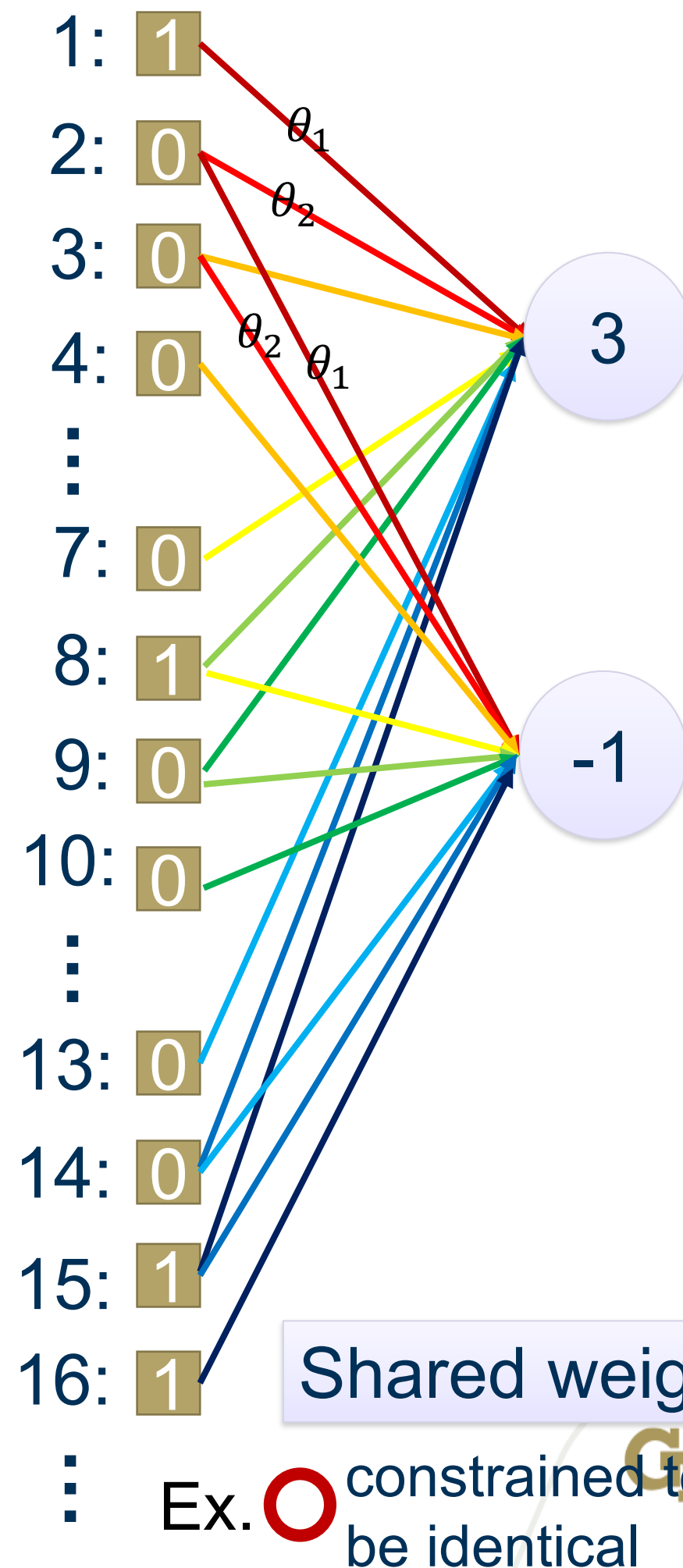
Only connect to 9 inputs, not fully connected



6 x 6 image

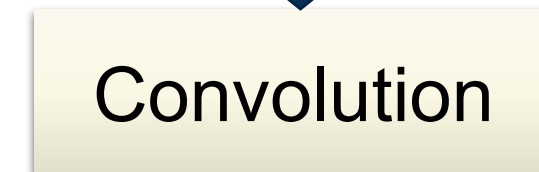
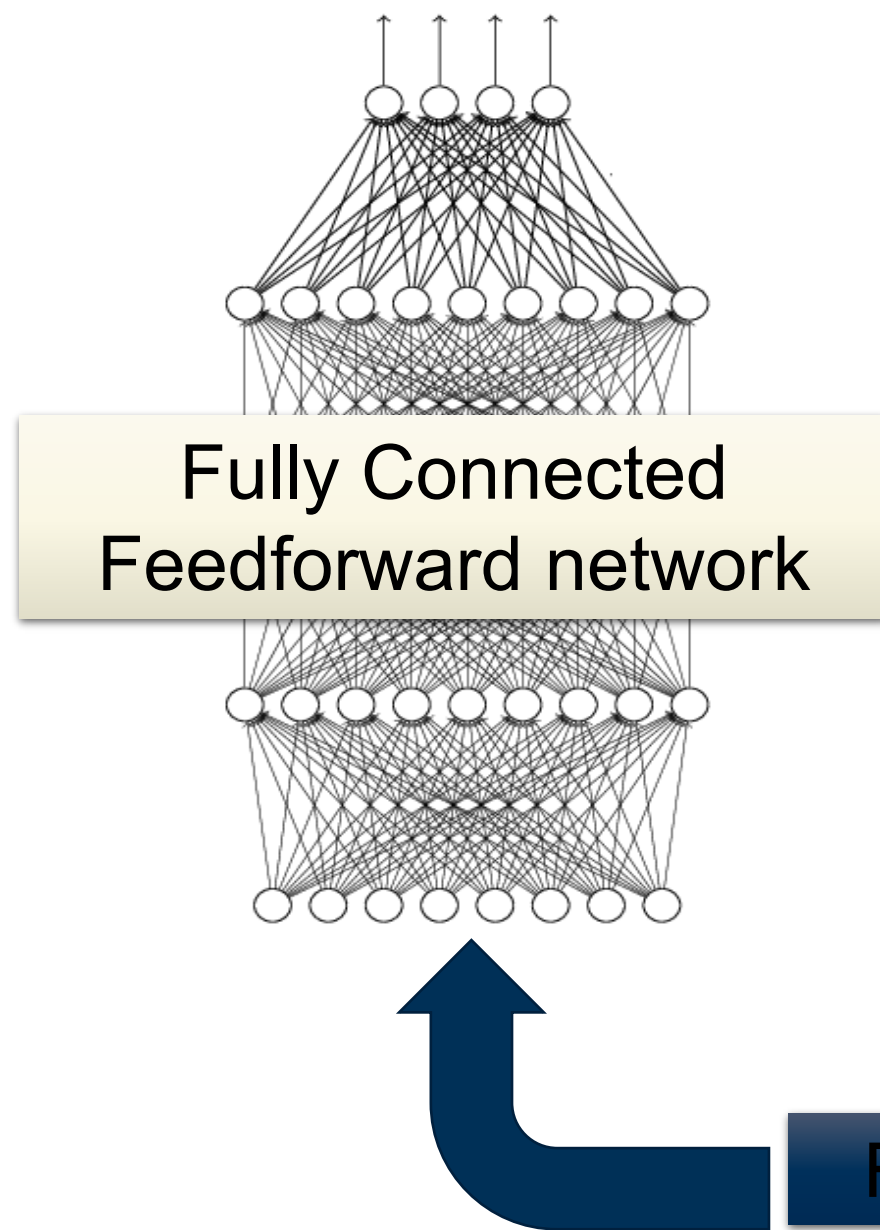
Fewer parameters

Even fewer parameters



The whole CNN

cat dog



Can repeat
many times

Flattened

Max Pooling

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

-1	-1	-1	-1
-1	-1	-2	1
-1	-1	-2	1
-1	0	-4	3

Why Pooling

- Subsampling pixels will not change the object

bird



Subsampling

bird



We can subsample the pixels to make image smaller

➡ fewer parameters to characterize the image

A CNN compresses a fully connected network in three ways:

- Reducing number of connections
- Shared weights on the edges
- Max pooling further reduces the complexity

Max Pooling

New image
but smaller

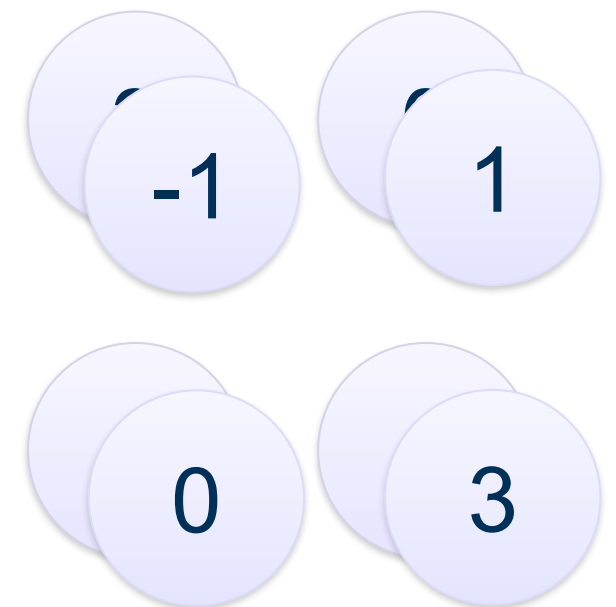
Each filter
is a channel

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

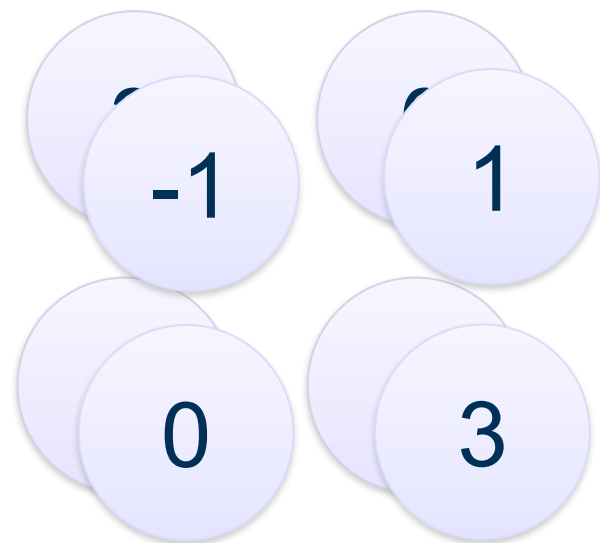
Conv

Max
Pooling



2 x 2 image

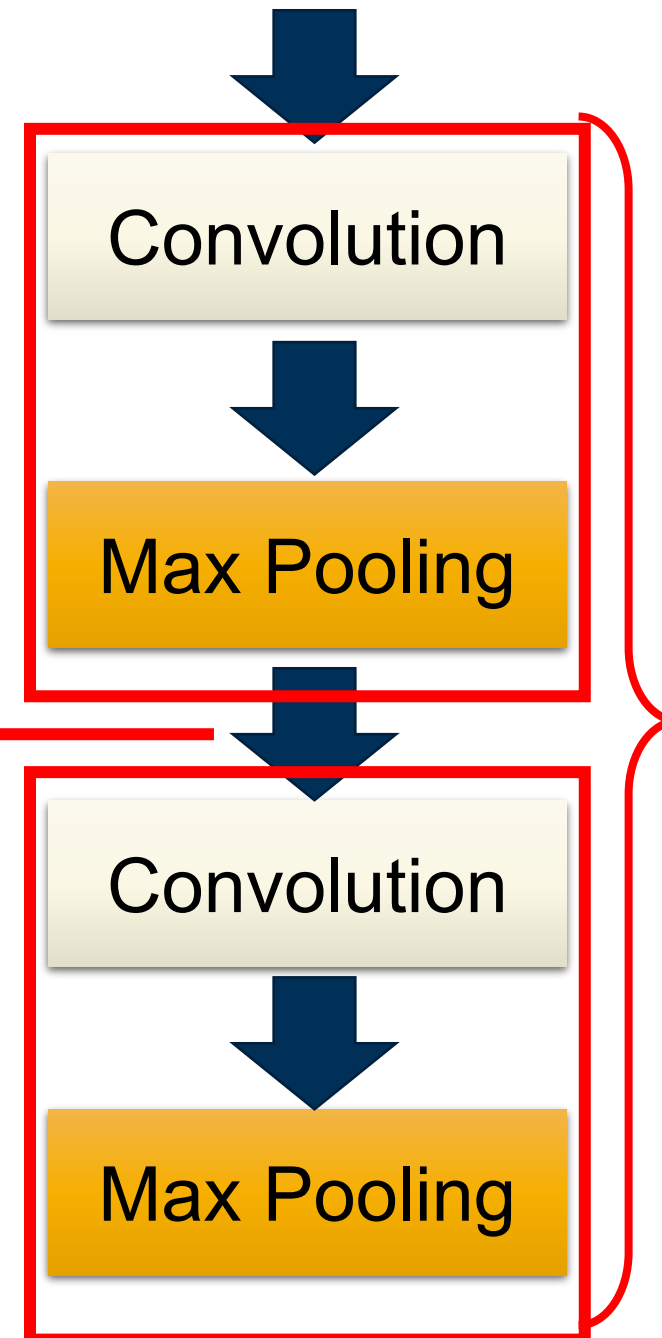
The whole CNN



A new image

Smaller than the original image

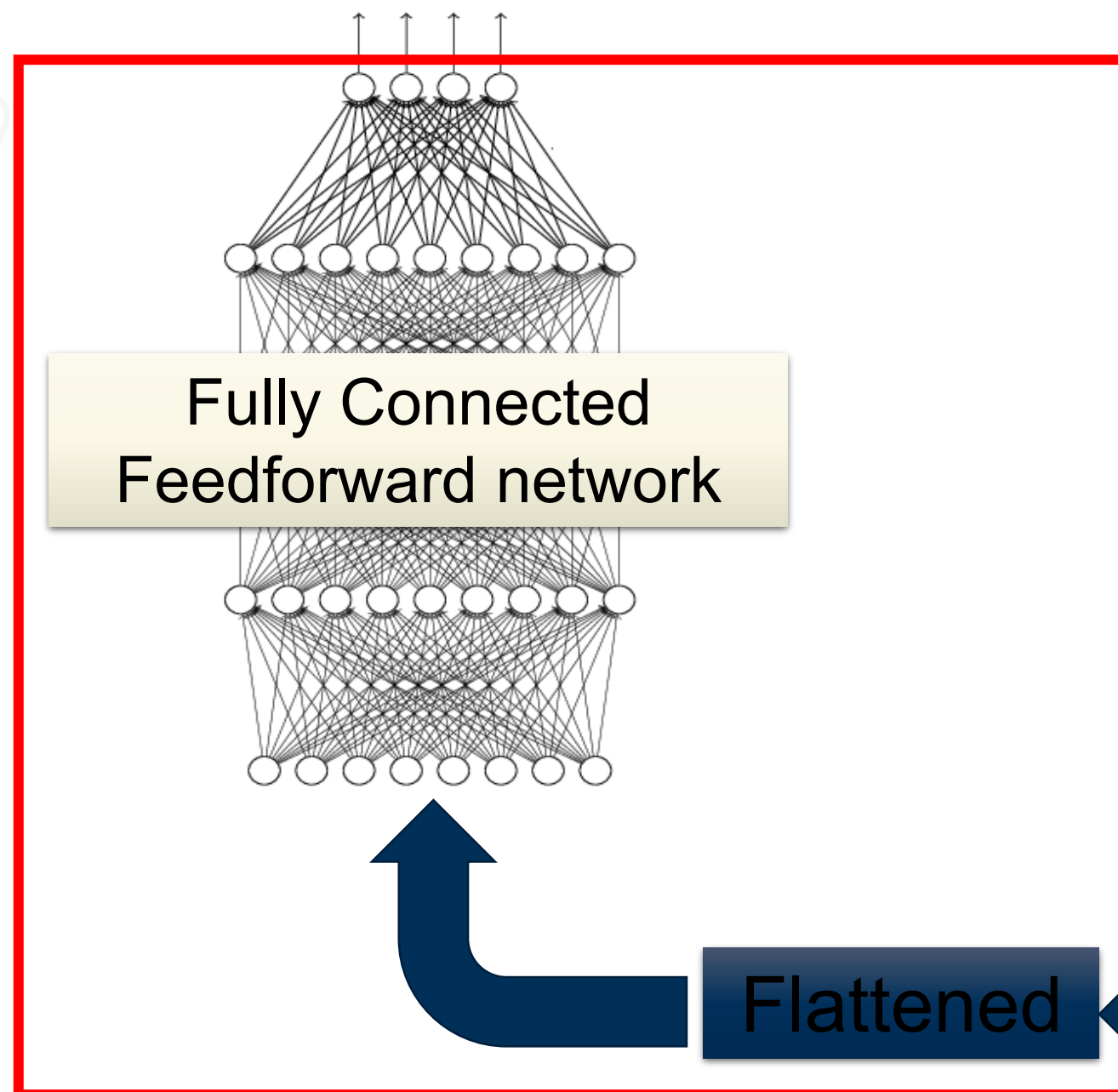
The number of channels is the number of filters



Can repeat many times

The whole CNN

cat dog



Convolution

Max Pooling

Convolution

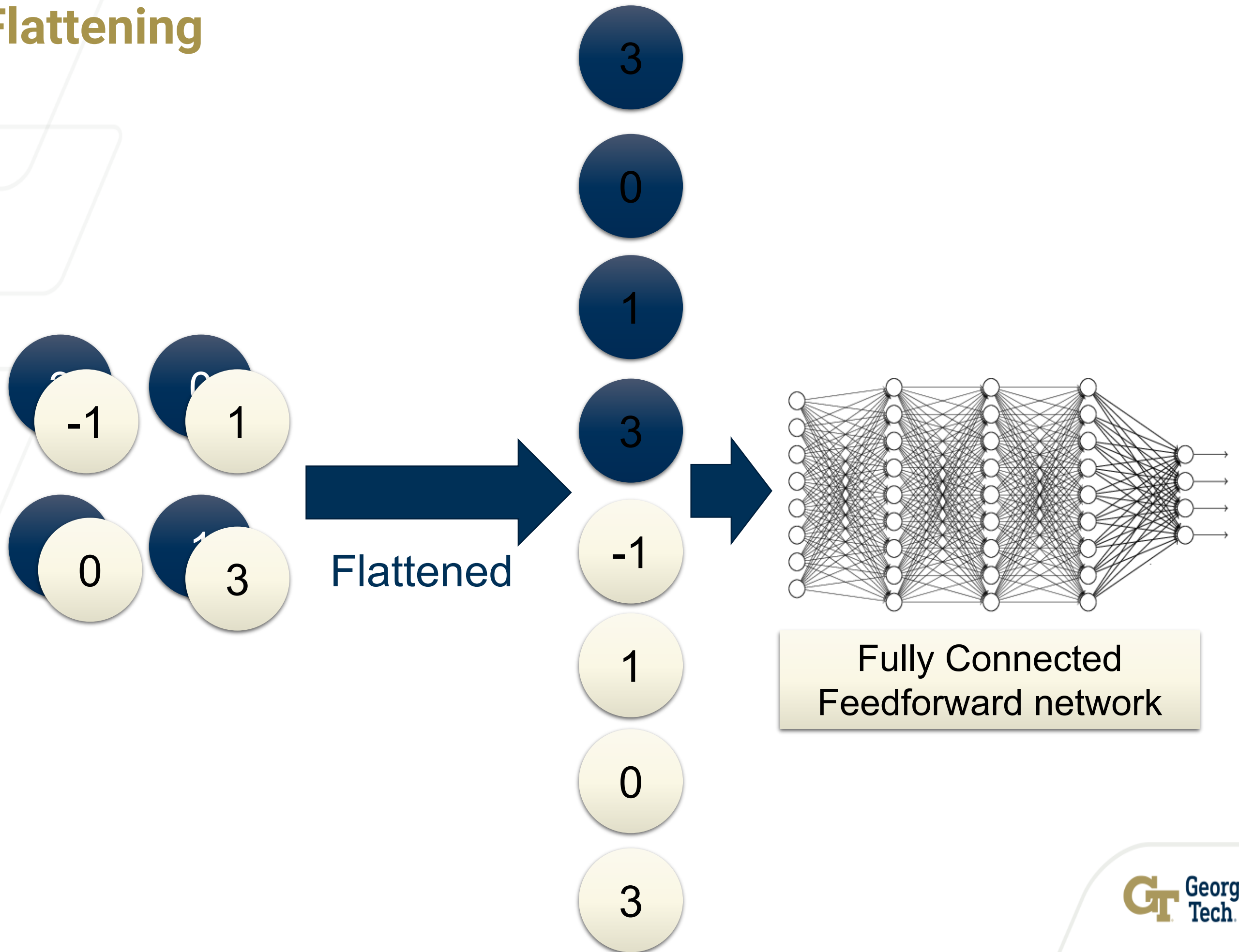
Max Pooling

A new image

A new image

Flattened

Flattening



CNN in Keras

Only modified the *network structure* and *input format* (vector -> 3-D tensor)

```
model2.add( Convolution2D( 25,3,3,  
                           input_shape=(28,28,1)) )
```

1	-1	-1	1	-1
-1	1	-1	1	-1
-1	-1	-1	1	-1
-1	-1	-1	1	-1

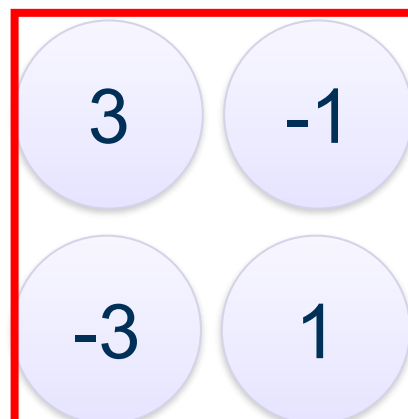
There are
25 3x3
filters.

Input_shape = (28 , 28 , 1)

28 x 28 pixels

1: black/white, 3: RGB

```
model2.add(MaxPooling2D((2,2)))
```



input

Convolution

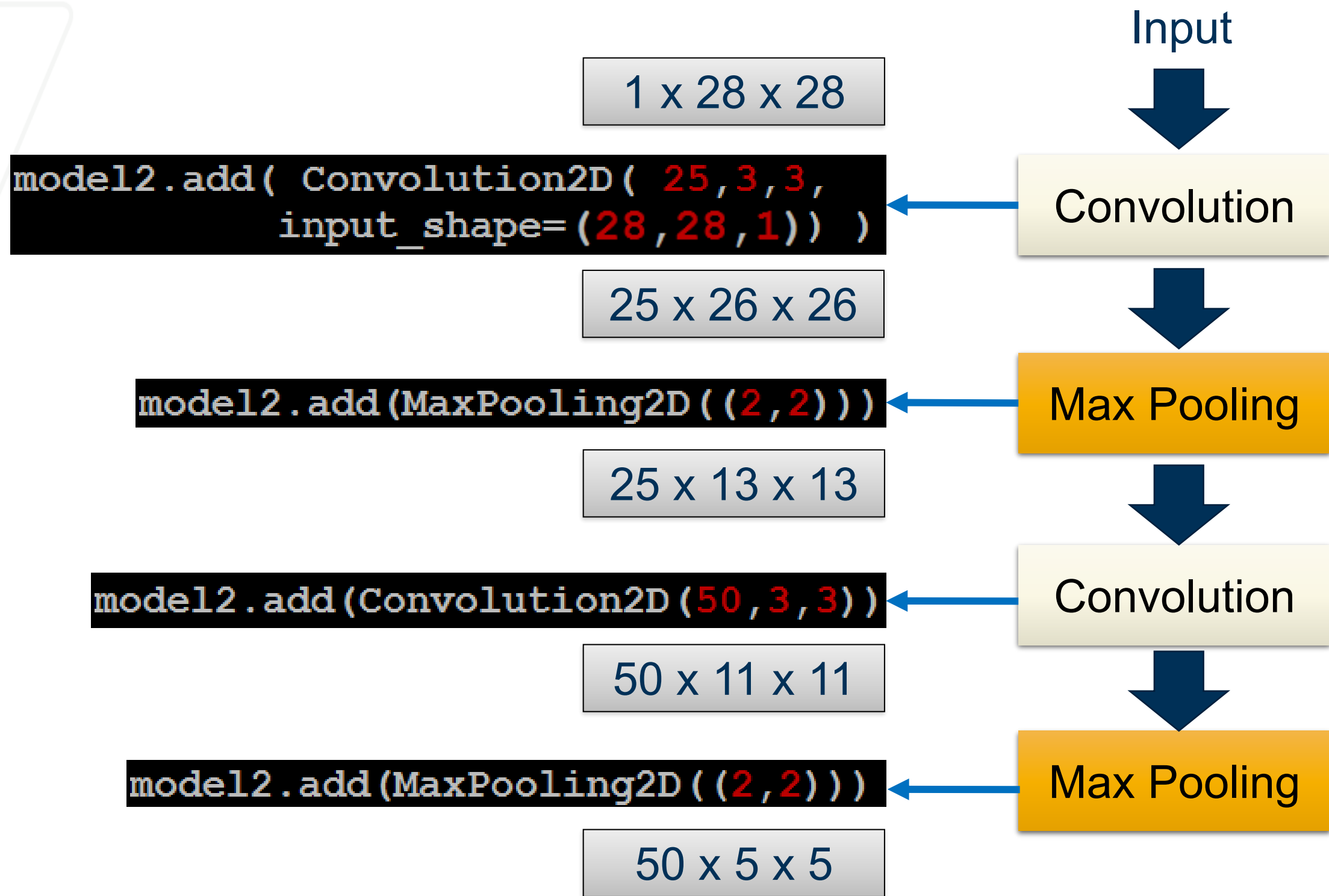
Max Pooling

Convolution

Max Pooling

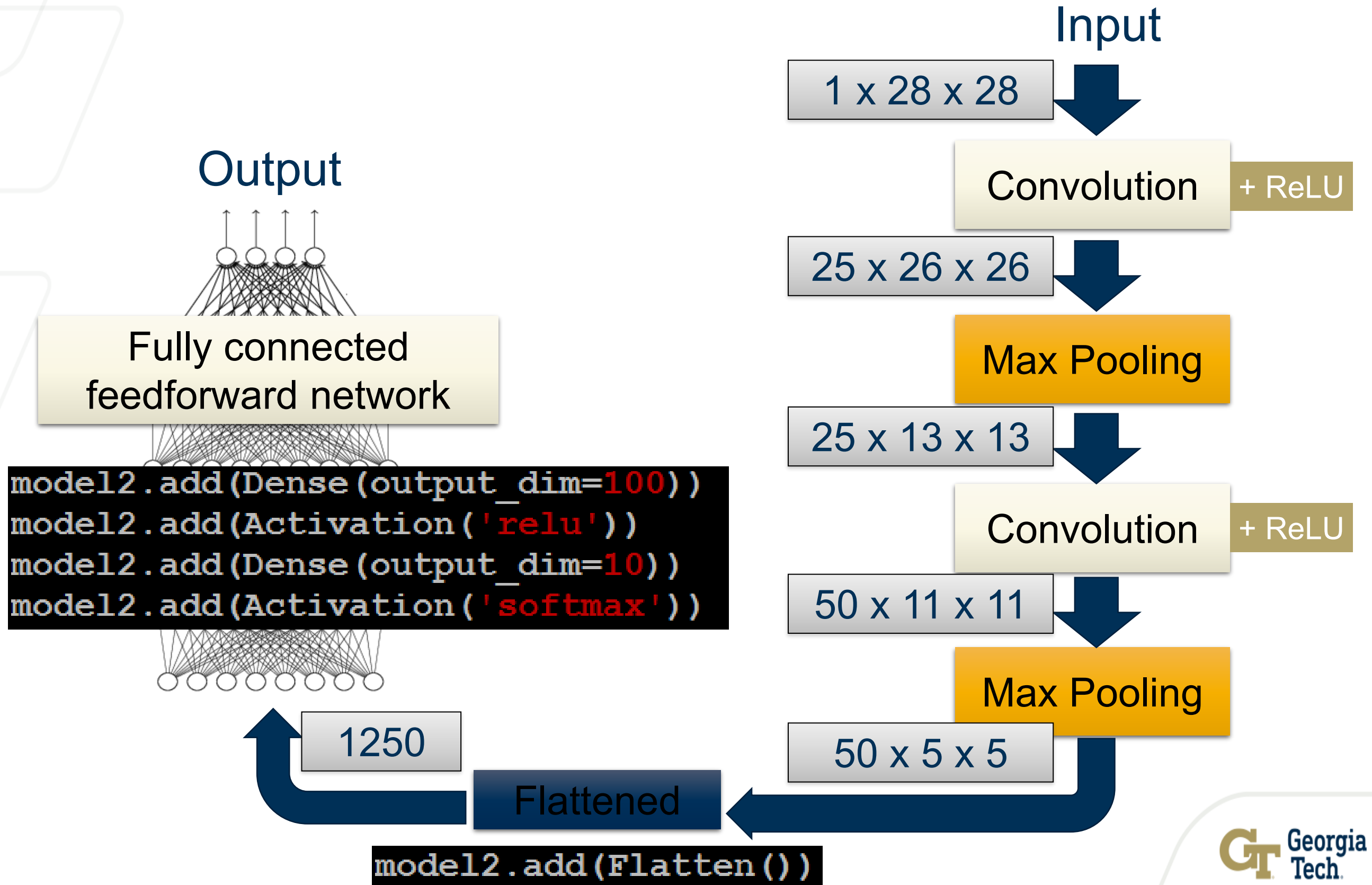
CNN in Keras

Only modified the *network structure* and *input format* (vector -> 3-D array)



CNN in Keras

Only modified the *network structure* and *input format* (vector -> 3-D array)



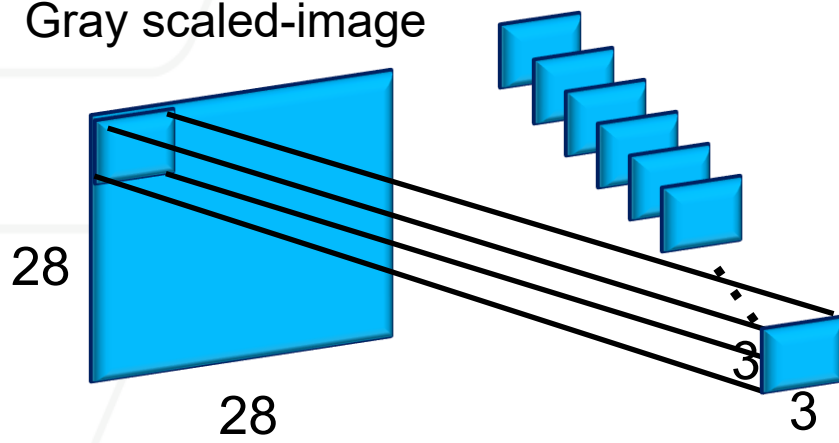
Number of Parameters

$25 \times 3 \times 3 + 25$ parameters

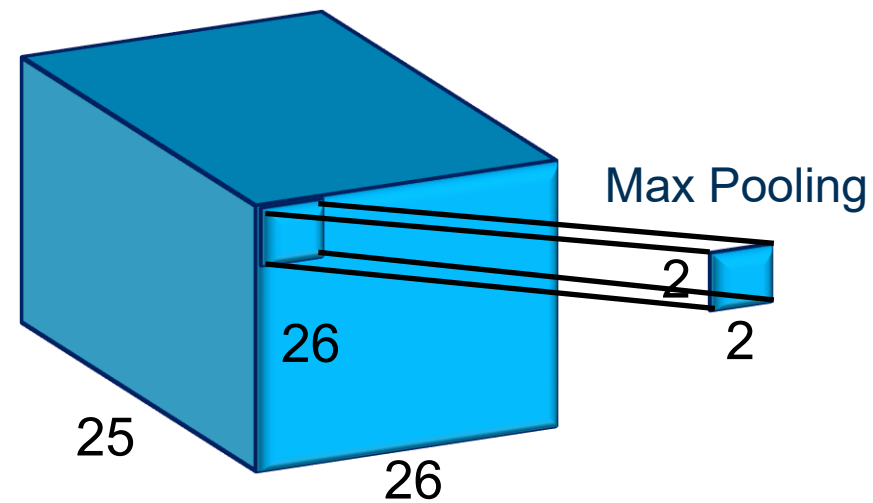
25 filters - Conv1

25: 3×3

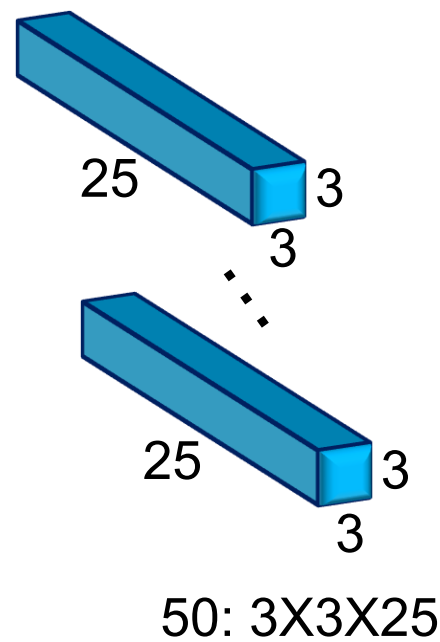
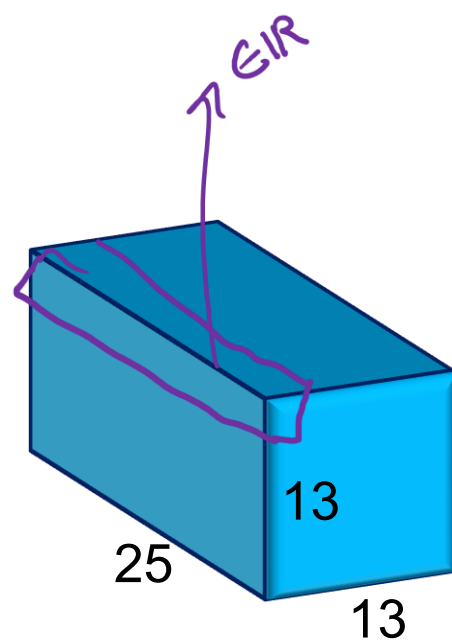
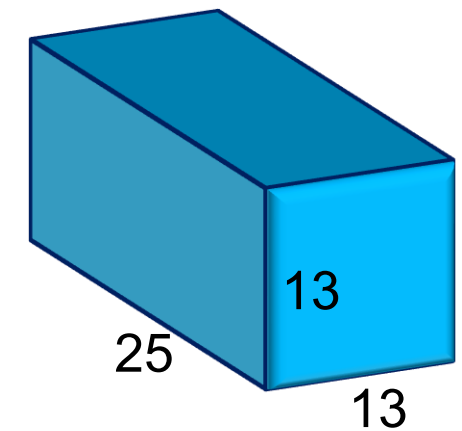
Gray scaled-image



Convolved result for 25 filters



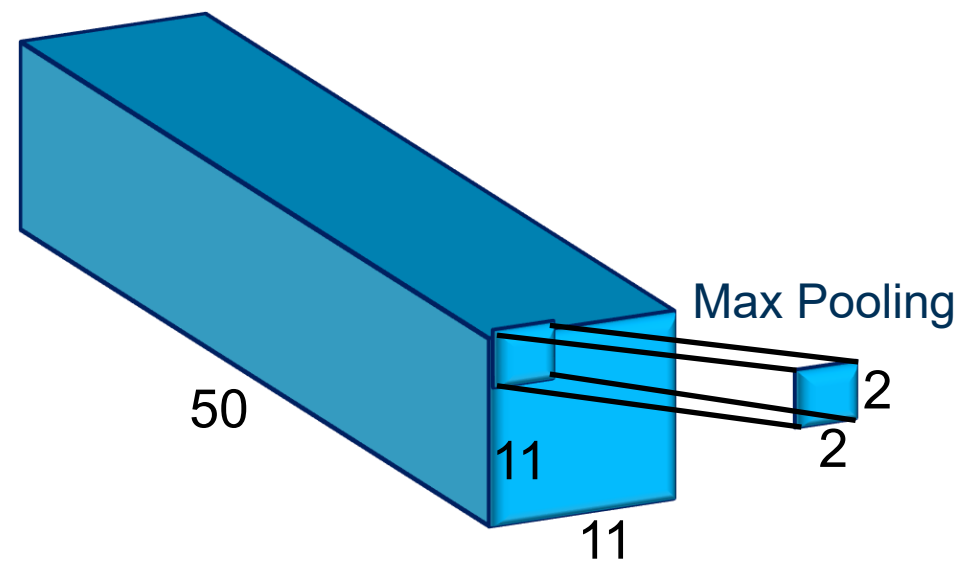
Result after Max Pooling



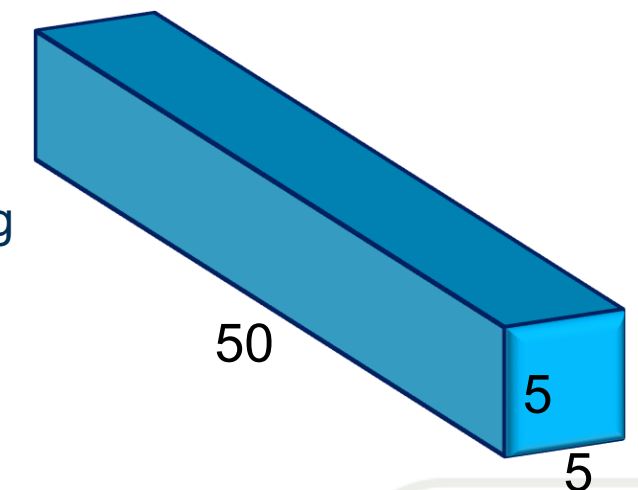
50 filters - Conv2

$50 \times 3 \times 3 \times 25 + 50$ parameters

Convolved result for 50 filters

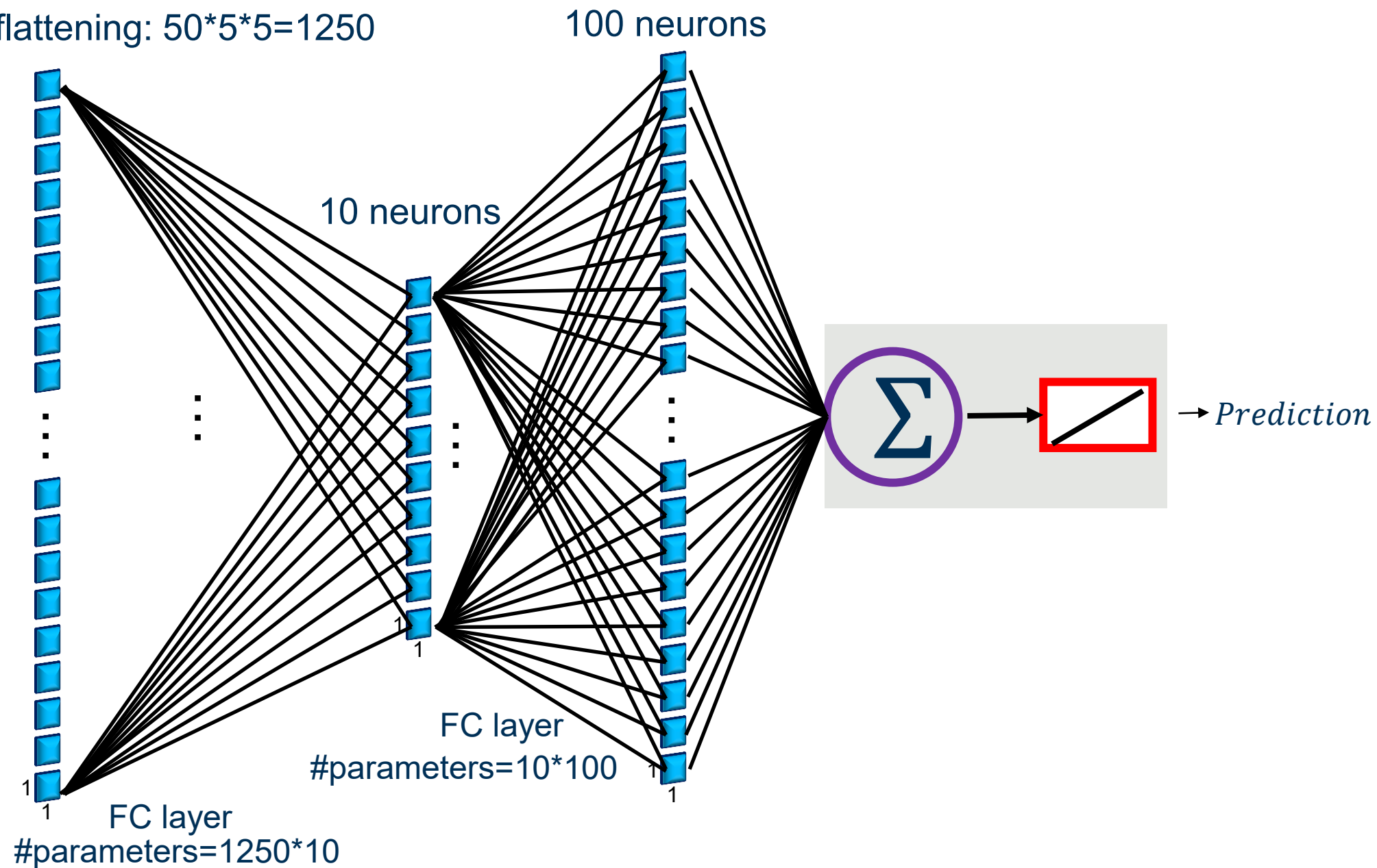
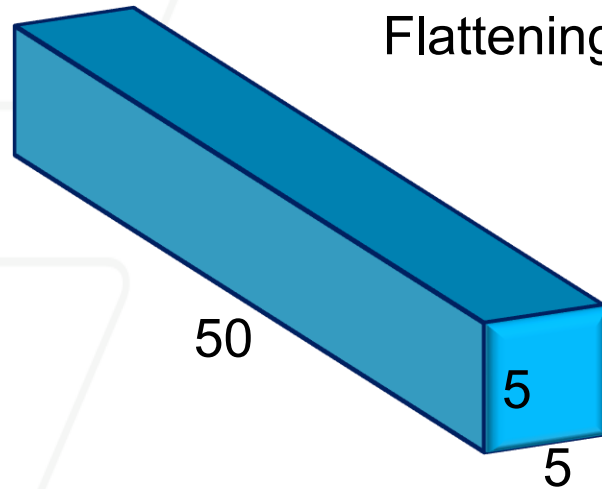


Result after Max Pooling



neurons after flattening: $50 \times 5 \times 5 = 1250$

Flattening



10 CNN Architecture